COMPLETE SYLLABUS AT A GLANCE

All 5 Sections: Topic Checklist

Every topic you need, on one page. Start with the gold (must-do) topics of each section. Tick **L** when you have learned the concept, **P** when you have practised it (20+ questions) and **M** when you have done it in a timed mock. Detailed notes for each section follow on the next pages.

● Must-do ● Important ● Good to know **L** = Learned **P** = Practised **M** = Mock done

Topic	L	P	M
S1 Numerical Ability (20 Q | 25 min)			
● Percentages	☐	☐	☐
● Profit & Loss	☐	☐	☐
● Time, Speed & Distance	☐	☐	☐
● Time & Work	☐	☐	☐
● Ratio & Proportion	☐	☐	☐
● Number System	☐	☐	☐
● Averages	☐	☐	☐
● LCM & HCF	☐	☐	☐
● SI & CI	☐	☐	☐
● Probability	☐	☐	☐
● Permutation & Combination	☐	☐	☐
● Mensuration	☐	☐	☐
● Mixtures & Alligation	☐	☐	☐
● Ages, Simplification, Approximation	☐	☐	☐
S2 Reasoning Ability (20 Q | 25 min)			
● Series (number and alphabet)	☐	☐	☐
● Coding – Decoding	☐	☐	☐
● Seating Arrangements	☐	☐	☐
● Syllogisms	☐	☐	☐
● Blood Relations	☐	☐	☐
● Directions & Distances	☐	☐	☐
● Clock & Calendar	☐	☐	☐
● Ranking & Ordering	☐	☐	☐
● Alphabet Test & Symbol Operations	☐	☐	☐
● Analogies & Odd One Out	☐	☐	☐

Topic	L	P	M
S3 Verbal Ability (25 + tasks Q | 25 min)			
● Sentence Completion: grammar	☐	☐	☐
● Passage Recall	☐	☐	☐
● Email Writing	☐	☐	☐
● Sentence Completion: vocabulary	☐	☐	☐
● Idioms, Phrases & Usage	☐	☐	☐
● Error Spotting & Parallelism	☐	☐	☐
● Typing Speed & Accuracy	☐	☐	☐
● Instruction Screens	☐	☐	☐
S4 Advanced Quant + Reasoning (14 Q | 25 min)			
● Data Interpretation	☐	☐	☐
● Analytical Puzzles	☐	☐	☐
● Data Sufficiency	☐	☐	☐
● Permutation & Combination (Adv.)	☐	☐	☐
● Set Theory & Venn Diagrams	☐	☐	☐
● Advanced Probability	☐	☐	☐
● Partnership	☐	☐	☐
● Critical Reasoning	☐	☐	☐
● Syllogisms (Advanced)	☐	☐	☐
S5 Advanced Coding (2 Q | 90 min)			
● Arrays	☐	☐	☐
● Strings	☐	☐	☐
● Math & Number Theory	☐	☐	☐
● Sorting & Searching	☐	☐	☐
● Programming Fundamentals	☐	☐	☐
● Hashing, Sets & Maps	☐	☐	☐
● Greedy & Dynamic Programming	☐	☐	☐
● Two Pointers & Sliding Window	☐	☐	☐
● Recursion & Backtracking	☐	☐	☐
● Stack & Queue	☐	☐	☐
● Matrix / 2D Arrays	☐	☐	☐
● Simulation & Real-world Problems	☐	☐	☐
● Linked List, Trees & Graphs	☐	☐	☐
● Bit Manipulation	☐	☐	☐

SECTION 1 OF 5 · PART A · FOUNDATION

Numerical Ability

20	25 min	1 min 15 s	15 of 20	No
Questions	Time	Per question	Target correct	Negative marking

Official syllabus: Number System | LCM & HCF | Percentages | Profit & Loss | Time, Speed & Distance | Time & Work | SI & CI | Averages | Ratio & Proportion | Probability | P & C | Mensuration | Ages | Mixtures

Topics and what to master

#	Topic	Priority	What to master
1	**Percentages**	Must-do	Fraction-to-percentage table, successive change, population and salary problems. Convert to fractions to calculate faster.
2	**Profit & Loss**	Must-do	CP, SP, MP, discount, successive discounts and false weights. Always take the percentage on the right base.
3	**Time, Speed & Distance**	Must-do	Average and relative speed, trains, boats and streams, races, meeting and chasing problems. Keep units consistent.
4	**Time & Work**	Must-do	Efficiency and total-work method, pipes and cisterns, alternate days, work and wages.
5	**Ratio & Proportion**	Must-do	Dividing amounts, compound ratio, direct and inverse proportion, partnership basics.
6	**Number System**	Must-do	Divisibility, remainders, unit digits, number of factors, base conversion and trailing zeros.
7	**Averages**	Important	Weighted average, change in average on adding or removing a value, age-based averages.
8	**LCM & HCF**	Important	Prime factorisation, bells ringing together, greatest number with a given remainder.
9	**SI & CI**	Important	SI and CI formulas, half-yearly and quarterly, difference between CI and SI for 2 and 3 years.
10	**Probability**	Important	Coins, dice, cards and bags; at least and at most; independent events.
11	**Permutation & Combination**	Important	nPr, nCr, arrangements with repetition, word rank, restricted selection.
12	**Mensuration**	Important	Area and perimeter of 2D shapes; volume and surface area of cube, cuboid, cylinder, cone and sphere.
13	**Mixtures & Alligation**	Important	Alligation rule, replacement of liquid, cost of mixtures.
14	**Ages, Simplification, Approximation**	Good to know	Age equations and ratios, BODMAS, squares and cubes, quick approximation.

Must-know rules

- Learn squares to 30, cubes to 15, tables to 20 and common fractions as percentages (1/7 = 14.28%).
- Successive percentage change: a + b + ab/100.
- Speed in km/h to m/s: multiply by 5/18. Reverse: multiply by 18/5.
- Work done = efficiency x time. Take LCM of days as total work.
- Profit% = (SP − CP)/CP x 100, always on cost price unless stated.

Common traps

- Taking percentage on the wrong base (CP vs SP, old vs new value).
- Forgetting to convert minutes to hours in speed questions.
- Spending over 2 minutes on one question. Skip and return.
- Rounding too early in TITA (typed answer) questions.
- Leaving questions blank. There is no negative marking.

SECTION 2 OF 5 · PART A · FOUNDATION

Reasoning Ability

20	25 min	1 min 15 s	16 of 20	No
Questions	Time	Per question	Target correct	Negative marking

Official syllabus: Series | Coding – Decoding | Blood Relations | Directions | Seating Arrangements | Syllogisms | Analogies | Odd One Out | Clock & Calendar | Ranking | Alphabet Test | Symbol Operations

Topics and what to master

#	Topic	Priority	What to master
1	**Series (number and alphabet)**	Must-do	Arithmetic, geometric, square, cube, alternating and mixed series; missing and wrong term; letter positions.
2	**Coding – Decoding**	Must-do	Letter shifts, reverse alphabet, number and symbol codes, conditional and mixed coding.
3	**Seating Arrangements**	Must-do	Linear (one and two rows), circular and square tables. Draw the layout first and place definite clues before conditional ones.
4	**Syllogisms**	Must-do	All, Some, No and Some-not; Venn diagrams; definite vs possibility; either-or conclusions.
5	**Blood Relations**	Must-do	Family trees with gender marks, coded relations, pointing to a photograph.
6	**Directions & Distances**	Important	Net displacement with Pythagoras, left and right turns, shadow-based directions.
7	**Clock & Calendar**	Important	Angle between hands, overlaps, gain and loss, day of the week using odd days, leap years.
8	**Ranking & Ordering**	Important	Position from left and right, total persons = rank from top + rank from bottom − 1, interchange of positions.
9	**Alphabet Test & Symbol Operations**	Important	Alphabet positions, meaningful words, letter counting, redefined operators and coded inequalities.
10	**Analogies & Odd One Out**	Good to know	Word, number and letter analogies; classification. Quick marks if you know the patterns.

Must-know rules

- Positions: A=1 ... Z=26, and opposite letter pairs add to 27.
- Total persons in a row = position from left + position from right − 1.
- Venn diagram for every syllogism, never intuition.
- Clock: angle = $|30H - 5.5M|$. Hands overlap 11 times in 12 hours.
- Odd days: ordinary year 1, leap year 2, century 0 or 1 (400 years = 0).

Common traps

- Mixing left and right in direction and seating questions.
- Treating 'some' as 'all' in syllogisms.
- Starting a puzzle set without a table or diagram.
- Missing the word 'definitely' in conclusions.
- Forgetting that facing direction flips left and right in circular seating.

SECTION 3 OF 5 · PART A · FOUNDATION (UPDATED 2027 PATTERN)

Verbal Ability

25+	25 min	25 s	9 min	3
Items + tasks	Time	Per sentence	Email writing	Task types

Official syllabus: Sentence Completion (20 items, 25 s each) | Passage Recall (4 + 4, 30 s read + 90 s write) | Email Writing (1, 9 min) | Instruction screens

Topics and what to master

#	Topic	Priority	What to master
1	**Sentence Completion: grammar**	Must-do	Tenses, subject-verb agreement, articles, prepositions, pronouns, conjunctions. Eliminate wrong options in seconds.
2	**Passage Recall**	Must-do	The passage disappears after 30 seconds. Note names, numbers and order of events, then rebuild in short clear sentences within 90 seconds.
3	**Email Writing**	Must-do	Subject, greeting, purpose, details, action, closing. Practise request, complaint, apology and follow-up emails in 9 minutes.
4	**Sentence Completion: vocabulary**	Important	Synonyms, antonyms, word choice and collocations. Learn 10 words daily.
5	**Idioms, Phrases & Usage**	Important	Common idioms, phrasal verbs, confusing word pairs.
6	**Error Spotting & Parallelism**	Important	Sentence structure, modifiers, parallel construction.
7	**Typing Speed & Accuracy**	Important	Recall and email are typed. Aim for 30+ words per minute with few errors.
8	**Instruction Screens**	Good to know	Short timed screens (25 s and 15 s). Read fast; the timer keeps running.

Must-know rules

- Each item has its own timer: 25 s for each sentence, 120 s for each recall task, 540 s for the email.
- Subject-verb agreement: find the true subject, ignore words in between.
- Email structure: Subject → Greeting → Purpose → Details → Action → Closing.
- In recall, write facts (who, what, when, number) first, then connect them.
- Check spelling and punctuation before the email timer ends.

Common traps

- Spending 40 s on one sentence-completion item.
- Trying to memorise the passage word for word.
- Writing a long email and running out of time.
- Informal tone or text-style spelling in emails.
- Slow typing that wastes the 90 s recall window.

SECTION 4 OF 5 · PART B · ADVANCED

Advanced Quant + Reasoning

14	25 min	1 min 47 s	9–10 of 14	No
Questions	Time	Per question	Target correct	Negative marking

Official syllabus: Data Interpretation | Set Theory & Venn Diagrams | Data Sufficiency | Syllogisms | Advanced Probability | Permutation & Combination | Partnership | Critical Reasoning | Analytical Puzzles

Topics and what to master

#	Topic	Priority	What to master
1	**Data Interpretation**	Must-do	Tables, bar, line, pie and mixed charts. Percentage change, share of total, ratios and averages. Approximate to save time.
2	**Analytical Puzzles**	Must-do	Multi-variable arrangement, scheduling, truth-and-liar and grouping puzzles. Build a table first.
3	**Data Sufficiency**	Must-do	Check Statement I alone, then II alone, then both together. Decide sufficiency, do not solve fully.
4	**Permutation & Combination (Adv.)**	Must-do	Restricted arrangements, distribution of objects, circular and digit-formation problems.
5	**Set Theory & Venn Diagrams**	Important	Two and three sets, only/exactly/at least counts, maximum and minimum overlap.
6	**Advanced Probability**	Important	Conditional probability, with and without replacement, arrangement-based probability.
7	**Partnership**	Important	Investment x time, entry and exit of partners, working partner with salary.
8	**Critical Reasoning**	Important	Assumption, inference, strengthen and weaken, course of action.
9	**Syllogisms (Advanced)**	Good to know	3–4 statement sets with possibility cases, solved with Venn diagrams.

Must-know rules

- Two sets: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- Three sets: add singles, subtract pairs, add the triple intersection.
- Data sufficiency order: I alone → II alone → both → neither.
- Approximate in DI: 48.7% of 623 ≈ 50% of 620.
- Start with the question group that has the most questions (a DI set or puzzle).

Common traps

- Solving the whole DI set before reading the questions.
- Calculating exactly when an approximation is enough.
- Getting stuck on one puzzle for over 4 minutes.
- Using both statements before testing each alone.
- Chasing all 14 questions instead of securing 9–10.

SECTION 5 OF 5 · PART B · ADVANCED

Advanced Coding

2	90 min	35 + 55	5	Hidden
Problems	Time	Easy + Medium (min)	Languages	Test cases

Official syllabus: Languages: C | C++ | Java | Python | Perl. Problem types: arrays | strings | math | sorting | recursion | DP and more. Scoring is by hidden test cases.

Topics and what to master

#	Topic	Priority	What to master
1	**Arrays**	Must-do	Traversal, rotation, prefix sums, Kadane, duplicates and missing numbers, rearrangement, frequency counting.
2	**Strings**	Must-do	Palindromes, anagrams, frequency, substrings, compression, reverse words, pattern matching.
3	**Math & Number Theory**	Must-do	Prime and sieve, GCD and LCM, Fibonacci, digit problems, modular arithmetic, base conversion.
4	**Sorting & Searching**	Must-do	Core sorts, binary search and its variants, custom comparators, k-th element.
5	**Programming Fundamentals**	Must-do	Input and output formats, multiple test cases, loops, patterns, data types and overflow.
6	**Hashing, Sets & Maps**	Must-do	Two-sum, group anagrams, first non-repeating element, subarray sum equals K.
7	**Greedy & Dynamic Programming**	Must-do	Coin change, knapsack, LIS, LCS, grid paths, subset sum. Usually the Medium problem.
8	**Two Pointers & Sliding Window**	Important	Pair sum, longest unique substring, fixed and variable windows.
9	**Recursion & Backtracking**	Important	Subsets, permutations, N-Queens, combination sum, maze paths.
10	**Stack & Queue**	Important	Balanced parentheses, next greater element, expression evaluation.
11	**Matrix / 2D Arrays**	Important	Spiral and diagonal traversal, rotation, transpose, search in sorted matrix.
12	**Simulation & Real-world Problems**	Important	Date and time, billing and parking logic, grid movement, scheduling.
13	**Linked List, Trees & Graphs**	Good to know	Reverse list, cycle detection, tree traversals, BFS and DFS basics.
14	**Bit Manipulation**	Good to know	Set bits, power of two, XOR tricks, toggling bits.

Must-know rules

- n up to 10^5 needs about O(n log n) or better. n up to 10^3 can use O(n^2).
- Write a brute-force solution first so you have partial marks, then optimise.
- Use 64-bit integers (long long, long) when sums or products can be large.
- Always test: empty input, one element, all equal, maximum size, negatives.
- Solve the Easy problem fully in about 35 minutes before the Medium.

Common traps

- Not reading constraints before choosing the approach.
- Missing multiple test cases or a wrong input and output format.
- Integer overflow and off-by-one errors.
- Spending 60 minutes on the Easy problem.
- Switching languages during preparation.